



continuous stirred tank reactor

A continuous stirred tank reactor (CSTR) is a vessel where reactants are fed continuously, the contents are well mixed, and products are continuously withdrawn so the reactor operates at steady state.^{[1] [2]}

Basic concept

- In a CSTR, fresh reactants flow in while an equal flow of reaction mixture leaves, keeping the liquid volume essentially constant.^[2]
- **Perfect** mixing is assumed, so composition and temperature are uniform throughout the tank and equal to those in the outlet stream.^{[3] [1]}

Main features

- A typical CSTR is a tank (often stainless steel) equipped with an impeller or agitator plus inlet and outlet ports for feeds and product.^{[4] [5]}
- Operating variables such as temperature, pressure, and residence time are controlled via heating/cooling systems and by adjusting flow rates.^{[6] [4]}

Advantages

- Continuous operation enables steady production without the repeated filling and emptying required in batch reactors.^{[7] [2]}
- Good mixing gives uniform temperature and concentration, helping maintain consistent product quality and improving heat removal for exothermic reactions.^{[4] [6] [7]}

Limitations

- A single CSTR has broad residence time distribution, so some fluid elements leave quickly while others stay longer, which can limit conversion for some reactions.^{[5] [4]}
- For higher effective residence time and less back-mixing, multiple CSTRs are often arranged in series to approximate plug flow behavior.^{[8] [4]}

Typical applications

- CSTRs are widely used in chemical and pharmaceutical manufacturing for liquid-phase reactions and processes involving slurries or catalysts.^{[8] [5] [4]}
- They also serve as bioreactors (e.g., for fermentation, hydrogen or biogas production) where continuous mixing keeps microorganisms suspended and ensures good mass

transfer.^[9] [4]

1. https://en.wikipedia.org/wiki/Continuous_stirred-tank_reactor
2. https://chem.libretexts.org/Courses/New_York_University/CHEM-UA_652:_Thermodynamics_and_Kinetics/01:_Lectures/1.23:_Continuously_stirred_tank_reactors
3. <https://www.cs.montana.edu/webworks/projects/stevesbook/contents/chapters/chapter008/section002/blue/page003.html>
4. <https://www.vapourtec.com/flow-chemistry/continuous-stirred-tank-reactor-cstr/>
5. <https://stolichem.com/chemical-process-technologies-continuous-stirred-tank-reactor/>
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